

Note that the smallest PIC is $\lambda_d(X; A)$ because $\mathbf{Q}$ has rank $(d + 1)$. We will show how $\lambda_d(X, A)$ plays an important role in understanding the feasibility of perfect concept erasure. Next, we introduce the concept of erasure-utility coefficient for a given joint distribution, $p_{A,X}$.

Definition A.3 (Erasure-Utility Coefficient). *The optimal erasure-utility coefficient for $p_{A,X}$ is:*

$$v^* := \inf \frac{I(Z; A)}{I(Z; X)}.$$

We observe that the erasure-utility coefficient is closely related to concept erasure, where we want to reduce $I(Z; A)$ while maximizing $I(Z; X)$. Under perfect concept erasure, $v^* = 0$. Next, we restate the result from [Du et al. \(2020\)](#), which provides an upper bound for the coefficient, v^* .

Lemma A.4 (Optimal Erasure-Utility Coefficient ([Calmon et al., 2017](#))). *Let $d := \min\{|\mathcal{A}|, |\mathcal{X}|\} - 1$, and $\lambda_d(A, X)$ be the smallest principal inertia component of any distribution $p(A, X)$ with finite support $\mathcal{A} \times \mathcal{X}$. Then,*

$$v^* \leq \begin{cases} \lambda_d(A, X), & \text{if } |\mathcal{X}| \leq |\mathcal{A}| \\ 0, & \text{otherwise} \end{cases}.$$

Lemma 3.2 (Perfect Concept Erasure Achievability). *Under the assumptions (A1)-(A4), for a joint distribution $p_{A,X}$ with finite support defined over $\mathcal{A} \times \mathcal{X}$, perfect concept erasure, $I(Z; A) = 0$, is achievable if either (a) smallest principal inertia component, $\lambda_d(A, X) = 0$ or (b) $|\mathcal{X}| > |\mathcal{A}|$.*

Proof of Lemma 3.2. From Lemma A.4 and Definition A.3, we can say that $v^* = 0$ when either $\lambda_d(A, X) = 0$ or $|\mathcal{X}| > |\mathcal{A}|$. $\square$

A.2 Proof of Lemma 3.3

Lemma 3.3 (Perfect Erasure Bound). *Under the assumptions (A1)-(A5), if an erasure function $f : \mathcal{X} \rightarrow \mathcal{Z}$ achieves perfect concept erasure $I(Z; A) = 0$, then the utility $I(Z; X)$ is bounded as:*

$$I(Z; X) = I(Z; X|A) \leq H(X|A),$$

where the equality is satisfied when $H(X|Z, A) = 0$.

Proof. The proof relies on the following chain rule for mutual information:

$$I(Z; X) = I(Z; A) + I(Z; X|A) - I(Z; A|X) \quad (9)$$

For completeness, we prove it below:

$$\begin{aligned} I(Z; A|X) &= H(Z|X) - H(Z|A, X) \\ \Rightarrow I(Z; A|X) &= H(Z) - H(Z) + H(Z|A) - H(Z|A) + H(Z|X) - H(Z|A, X) \\ \Rightarrow I(Z; A|X) &= [H(Z) - H(Z|A)] - [H(Z) - H(Z|X)] + [H(Z|A) - H(Z|A, X)] \\ \Rightarrow I(Z; A|X) &= I(Z; A) - I(Z; X) + I(Z; X|A) \\ \Rightarrow I(Z; X) &= I(Z; A) - I(Z; A|X) + I(Z; X|A). \end{aligned}$$

In the above result, we plug-in $I(Z; A) = 0$ (due to perfect erasure) and $I(Z; A|X) = 0$ (due to Markov assumption (A1)). We obtain the following result:

$$I(Z; X) = I(Z; X|A). \quad (10)$$

Now, replacing the upper bound of $I(Z; X|A) \leq H(X|A)$, we get:

$$I(Z; X) \leq H(X|A).$$

The equality is satisfied when the following holds:

$$\begin{aligned} I(Z; X|A) &= H(X|A) \\ \Rightarrow H(X|A) - H(X|Z, A) &= H(X|A) \\ \Rightarrow H(X|Z, A) &= 0. \end{aligned}$$

This completes the proof. $\square$

A.3 Justification for Data Constraints in Definition 4.1

In this section, we provide the justification behind the choice of data constraints necessary for optimal erasure, where the distribution of representations should be permutations. First, we present an additional Lemma that would be useful in developing the intuition behind the data constraints.

Lemma A.5 (Zero Conditional Entropy). *If X and Y are random variables defined on finite supports $\mathcal{X}$ and $\mathcal{Y}$ respectively and $Y = f(X)$, then $H(X|Y) = 0$ if and only if f is a bijective map.*

Proof of Lemma A.5. First, we prove the forward direction, which says that f is bijective given $H(X|Y) = 0$.

$$\begin{aligned} H(X|Y) &= 0 \\ \Rightarrow \sum_{y \in \mathcal{Y}} p(y) H(X|Y = y) &= 0 \\ \stackrel{(a)}{\implies} \forall y \in \mathcal{Y}, H(X|Y = y) &= 0 \end{aligned} \tag{11}$$

(a) holds because $\forall y \in \mathcal{Y}$, $p(y) > 0$. Eq. 11 implies that the probability distribution $\forall y, P(X|Y = y)$ is degenerate and there exists $x \in \mathcal{X}$ such that

$$\forall y, \exists x \text{ s.t. } P(X = x|Y = y) = 1, \forall x' \neq x, P(X = x'|Y = y) = 0. \tag{12}$$

Recall that we have $Y = f(X)$, and we want to show that f is invertible (and therefore bijective). Suppose not, then $\exists x, x' \in \mathcal{X}$ that $f(x) = f(x')$. This contradicts Eq. 12 because $P(x|y) = 1$. Therefore, f must be invertible and bijective.

Next, we prove the backward direction, which shows that $H(X|Y) = 0$ if f is bijective.

When f is bijective, the following conditional distribution holds for a given y :

$$P(X = f^{-1}(y)|Y = y) = 1 \text{ and } P(X = x|Y = y) = 0, \forall x \neq f^{-1}(y). \tag{13}$$

For a fixed y , using Eq. 13, we can write the conditional entropy as:

$$\begin{aligned} H(X|Y = y) &= - \sum_{x \in \mathcal{X}} P(X = x|Y = y) \log P(X = x|Y = y) \\ &= \left[1 \cdot \log 1 + \sum_{x \neq f^{-1}(y)} 0 \cdot \log 0 \right] \\ &= 0. \end{aligned} \tag{14}$$

Using the above result, we can write the following:

$$H(X|Y) = \sum_{y \in \mathcal{Y}} p(y) H(X|Y = y) = \sum_{y \in \mathcal{Y}} p(y) \cdot 0 = 0. \tag{15}$$

This completes the proof. $\square$

With the above result, we are prepared to explain the data constraints in Definition 4.1.

Definition 4.1 (Data Constraints). *We consider the class of data for which the distributions of different groups are permutations of each other:*

$$\forall(i, j), (a) |\mathcal{X}_i| = |\mathcal{X}_j|, (b) \forall x \in \mathcal{X}_i, P_i(x) = P_j(\sigma_{ij}(x)),$$

where $P_i = P(X|A = a_i)$, and for which there exists an erasure function f that achieves $I(Z; A) = 0$ and $I(X; Z) = H(X|A)$.

The intuition of this definition is that we want to find an f that makes use of the bijective maps between

First, we note that due to the disjoint support assumption (A4), we can write f as a piecewise function:

$$f(x) = \{f_i(x) | x \in \mathcal{X}_i\}, \text{ where } f_i : \mathcal{X}_i \rightarrow \mathcal{Z}. \quad (16)$$

From Lemma 3.3, we know that the utility outer bound holds when the following condition is true:

$$\begin{aligned} H(X|Z, A) &= 0 \\ \stackrel{(a)}{\implies} H(X|Z, A = a_i) &= 0, \forall i \in \{1, \dots, |\mathcal{A}|\} \end{aligned} \quad (17)$$

(a) holds because $\forall i \in \{1, \dots, |\mathcal{A}|\}, p(a_i) > 0$ by definition of the support. Applying Lemma A.5, we observe that Eq. 17 is satisfied only when all f_i 's are bijective maps.

The bijective maps f_i 's imply that the support size of the input and output must be the same $|\mathcal{X}_i| = |\mathcal{Z}|$ (the support of $\mathcal{X}$ and $\mathcal{Z}$ is finite as stated in Assumption (A2)). This implies for every pair of concept group $\forall(i, j), |\mathcal{X}_i| = |\mathcal{X}_j|$, which proves the first data constraint.

The bijective maps f_i 's also imply that $\forall i$ the distributions $P(Z|A = a_i)$ and $P(X|A = a_i)$ are permutations of each other, as shown below:

$$\begin{aligned} \forall i, P[Z = z|A = a_i] &= \sum_{x \in \mathcal{X}_i} \mathbb{1}[f_i(x) = z] P[X = x|A = a_i] \\ &= P_i[X = f_i^{-1}(z)] \\ &= P_i[X = \sigma_i(x)], \end{aligned} \quad (18)$$

where $\sigma_i(\cdot)$ is a permutation function and $P_i(X) = P(X|A = a_i)$.

Next, we focus on the condition of perfect erasure:

$$\begin{aligned} I(Z; A) &= 0 \\ \Rightarrow H(Z) - H(Z|A) &= 0 \\ \Rightarrow H(Z) &= H(Z|A) \\ \Rightarrow H(Z) &= H(Z|A = a_i), \forall i \in \{1, \dots, |\mathcal{A}|\}. \end{aligned} \quad (19)$$

Next, we note that the output distribution can be written as: $P(Z) = \sum_i p(a_i)P(Z|A = a_i)$. We can write down the entropy of Z as the entropy of the underlying distributions as below:

$$\begin{aligned} H(Z) &= H\left(\sum_i p(a_i)P[Z|A = a_i]\right) \\ &\geq \sum_i p(a_i)H(Z|A = a_i). \end{aligned} \quad (20)$$

Eq. 20 holds due to the concavity of the entropy function. However, for the Eq. 19 to hold, the equality in Eq. 20 needs to be satisfied. The equality is satisfied only when all the underlying distributions are equal, as shown below:

$$\begin{aligned} P[Z|A = a_i] &= P[Z|A = a_j], \forall(i, j) \\ \Rightarrow P_i[X = \sigma_i(x)] &= P_j[X = \sigma_j(x)] \end{aligned} \quad (21)$$

$$\begin{aligned} \Rightarrow P_i[X = x] &= P_j[X = \sigma_i^{-1}\sigma_j(x)] \\ \Rightarrow P_i[X = x] &= P_j[X = \sigma_{ij}(x)], \end{aligned} \quad (22)$$

where $\sigma_{ij} = \sigma_i \circ \sigma_j$. Eq. 21 holds due to the result in Eq. 18. The result in Eq. 22 implies that for all pairs of concept groups (i, j) , the underlying distributions are permutations of each other. This explains the second data constraint.

A.4 Justification for Method 4.2

In this section, we will show that the perfect erasure function defined in Eq. 3 achieves the MI outer bounds.

Method 4.2 (Perfect Erasure Function). *If the data constraints in Definition 4.1 hold, then define erasure function f as:*

$$f(x) = \{\sigma_i(x) \mid x \in \mathcal{X}_i, \sigma_i \in \Sigma_i\}, \quad (3)$$

where σ_i is any bijective function that transforms P_i into Q defined by the set, Σ_i below:

$$\Sigma_i = \{\sigma : \mathcal{X}_i \rightarrow \mathcal{Z} \mid \forall x \in \mathcal{X}_i, P_i(x) = Q(\sigma(x))\}. \quad (4)$$

First, we show that the perfect erasure function defined in Eq. 3 achieves the utility outer bound. Using Lemma 3.3, we get the following:

$$\begin{aligned} I(Z; X) &= I(Z; X|A) \\ &= \sum_i p(a_i) I(Z; X|A = a_i) \\ &= \sum_i p(a_i) [H(X|A = a_i) - H(X|Z, A = a_i)] \\ &= \sum_i p(a_i) H(X|A = a_i) \\ &= H(X|A). \end{aligned} \quad (23)$$

Eq. 23 can hold since $H(X|Z, A = a_i) = 0$ if $f_i : \mathcal{X}_i \rightarrow \mathcal{Z}$ is a bijective map.

Next, we show that the erasure function achieves perfect concept erasure. We compute the conditional distribution:

$$\forall z \in \mathcal{Z}, P(Z = z|A = a_i) = \sum_{x \in \mathcal{X}_i} \mathbf{1}[f_i(x) = z] P(X = x|A = a_i) \quad (24)$$

$$\begin{aligned} &= P_i(X = f_i^{-1}(z)) \\ &= Q(z). \end{aligned} \quad (25)$$

Eq. 24 holds because f_i is a bijective map. Eq. 25 holds because of the erasure function definition in Eq. 4, where $P(X = x|A = a_i) = P_i(x) = Q(f_i(x))$. This implies the probability distribution over $\mathcal{Z}$ is:

$$\begin{aligned} \forall z \in \mathcal{Z}, P(Z = z) &= \sum_i p(a_i) P[Z = z|A = a_i] \\ &= \sum_i p(a_i) Q(z) \\ &= Q(z). \end{aligned} \quad (26)$$

Next, we use Eq. 25 and Eq. 26 to derive the privacy mutual information as shown below:

$$\begin{aligned} I(Z; A) &= H(Z) - H(Z|A) \\ &= H(Z) - \sum_i p(a_i) H(Z|A = a_i) \\ &= H(Q) - \sum_i p(a_i) H(Q) \\ &= H(Q) - H(Q) \\ &= 0. \end{aligned} \quad (27)$$

Eq. 27 holds because $P(Z = z) = Q(z)$ and $\forall i \in \{1, \dots, |\mathcal{A}|\}, P(Z = z|A = a_i) = Q(z)$.

The idea of this is to find the erasure function defined in Method 4.2, which achieves the outer bounds $I(Z; X) = H(X|A)$ and $I(Z; A) = 0$.

B METHOD

B.1 Optimization Procedure

In this section, we present an alternative method to optimize the objective function presented in Eq. 7, when the input distributions of the concept groups are unequal. This optimization is dependent on two sets of variables: common support Q and coupling maps $\Gamma^{(i)} = \Gamma(P_i, Q)$. The key idea is to eliminate one set of variables and write the objective function only in terms of the coupling maps, since $Q = \sum_k \Gamma_{kj}^{(i)}$ is equal to the marginals of the coupling maps.

Therefore, we can write the objective as shown below:

$$\max_{\{\Gamma^{(i)}\}_{i=1}^{|\mathcal{A}|}} \frac{1}{|\mathcal{A}|} H\left(\sum_k \Gamma_{kj}^{(i)}\right) - \sum_i p(a_i) H_{\Gamma^{(i)}}\left(P_i, \sum_k \Gamma_{kj}^{(i)}\right), \quad (28)$$

$$\text{where } \forall (i_1, i_2) \in [1, |\mathcal{A}|], \sum_k \Gamma_{kj}^{(i_1)} - \sum_k \Gamma_{kj}^{(i_2)} = 0 \text{ and } \sum_k \Gamma_{kj}^{(i)} = P_i. \quad (29)$$

The above optimization can be solved using projected gradient descent (Madry et al., 2017), where we take unconstrained gradient steps using the differentiable objective in Eq. 28 and then projecting the solutions obtained, $\{\Gamma_i\}$ to the linear constraints in Eq. 29. However, since Eq. 28 is non-convex, solving this optimization still doesn't guarantee convergence to the global optima.

We applied this optimization method in small-scale synthetic setups and found it to function well. However, this method is not scalable to probability distributions with a large support set. This is because the number of linear constraints in Eq. 29 blows up, and the projection step is extremely expensive in practice.